Toy Universe Overview

A stipulated amplitude-record world, and a scientific case against treating it as ours

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Contents

The one-page account **1**

Why this candidate, rather than its competitors? **2**

What exists *inside* this mathematical world? **2**

Three internal demonstrations **3**
1. Interference with and without records 3
2. A classical-looking probability transition 3
3. A growing ring 3

What would have to be true for this to matter scientifically? **4**

Why the mathematics cannot be stretched into cosmology **4**

The seven Millennium Problems are not downstream of this toy **4**

The outcome of the adversarial sub-PCPs **5**

Closing judgment **5**

Grounding references **6**

> **Interpretation notice.** This is an invented, finite, mathematically specified toy universe. It is **not** an experimentally supported ontology, a derivation of quantum or classical mechanics from prephysical premises, a theory of quantum gravity, or a solution to any Millennium Prize Problem. The user-directed research program remains in **Parent Unary PCP: Initial External Grounding, OPEN**. This is a session-ending hypothetical artifact for future scrutiny, not a phase transition. Researchers should give its physical claims **very low prior weight**.

The one-page account

Imagine a miniature world built from a circular row of sites. Each site carries a finite set of possible mathematical configurations. The whole row also has a single size variable. At each **formal update index**, the configurations evolve under a chosen reversible amplitude rule; a record-making operation may store some information and suppress interference; and the size variable may increase under a separately chosen expansion rule.

Unlike a cellular automaton that stores only definite bits, this world is permitted to use complex amplitudes. Unlike a conventional quantum theory presented as an explanatory achievement, this construction **admits openly that complex amplitudes, tensor products, the probability rule,**

and update steps are installed as axioms. The structure can demonstrate interference and a classical-looking probability limit, but it does not explain why nature would choose those axioms.

The model is called **FAR-1: Finite Amplitude-Record Universe**, an explicitly stipulated replacement *test specimen*, **not** a replacement selected ontology. Its purpose is to offer a complete target to attack: a small world with state variables, update rules, records, observables, a simple expansion register, and reproducible examples. If it proves sterile, FAR-1 can be discarded without losing any previously supported mathematics.

The earlier TO-0 contextual toy remains withdrawn as a total-ontology candidate. It is retained only as a negative control. FAR-1 does not repair TO-0 by explaining its missing quantum selection; instead, FAR-1 changes the task and **imports quantum-like structure to build an internally executable example**. That change must not be hidden.

Why this candidate, rather than its competitors?

Four provisional constructions were compared with bounded adversarial reviews:

Construction	What it can do	Decisive limitation	Outcome
Finite classical layers	Probabilities, stationary distributions, finite records	Cannot generate general interference while keeping only classical affine probability rules	Retain as countermodel
Unrestricted contextual tables, including TO-0	Local, quantum-compatible and PR-type output tables	Does not select the quantum subset or the underlying mechanism	Ontological candidacy withdrawn
Bare causal/incidence graph	Order, compatibility, local counting	Does not select an operational probability theory, metric or clock	Retain as comparator
FAR-1 amplitude + record + size register	Consistent finite dynamics, interference, observation records, a toy expansion history	All central physical structures are imported; no independent evidence	**Selected for stipulation only**

This is not a ranking of plausible theories of nature. FAR-1 wins only a deliberately limited design contest: it is the most **explicitly executable and falsifiable as a mathematical artifact** within this session's resources.

What exists *inside* this mathematical world?

There are N sites arranged in a ring. Each site has two basis labels, 0 and 1. The complete mathematical matter state is a density operator acting on a 2^N-dimensional complex vector space. The labels are **not claimed to be physically emergent qubits**; they are stipulated two-level variables.

The ring has a positive scale a. Adjacent site labels are coupled, and every site can flip under a local field. The coupling strength can vary as $1/a$. A recording rule sometimes records a basis label and sometimes remains silent. The record operation causes dephasing when its outcome is ignored. The expansion rule changes a using positive terms intended to resemble radiation, matter, and a cosmological constant, with an optional toy backreaction from the matter configuration.

Five choices are therefore doing most of the explanatory work: the complex state space; the update generator; the Born-style record rule; the choice of a formal update index; and the Friedmann-

inspired size update. **None of these five is derived from a prephysical noise-like or incidence-only structure.**

The model distinguishes a *mathematical ontology*, where $(a, \rho, \text{record history})$ defines what the toy world contains, from a *physical ontology*, which would need independently tested bridges to actual preparations, clocks, distances and measurements. Only the former is given here.

Three internal demonstrations

1. Interference with and without records

In the single-site module, two reversible mixing operations surround an adjustable relative phase. With no intervening record, the probability of outcome 0 is

$$p_0 = \frac{1 + \cos\phi}{2}.$$

If the record operation dephases the intermediate state with strength $\kappa \in [0, 1]$, the same output becomes

$$p_0 = \frac{1 + (1 - \kappa)\cos\phi}{2}.$$

Phase	No record $\kappa = 0$	Partial record $\kappa = 1/4$	Complete record $\kappa = 1$
0	1.000	0.875	0.500
$\pi/2$	0.500	0.500	0.500
π	0.000	0.125	0.500

These numbers were independently recomputed with a finite density-matrix implementation. They are not measurements. Crucially, the Born probability and the mixing transformation were **put into** the model. The result is an internally correct demonstration, not quantum emergence.

2. A classical-looking probability transition

If recording completely destroys coherence in the chosen basis, a finite unitary transformation induces a stochastic transition on diagonal probabilities:

$$T_{ji} = |U_{ji}|^2.$$

For a two-level Hadamard mixing operation, T has all entries $1/2$. For a permutation operation, the transition is deterministic. These are legitimate classical probability dynamics **within the toy**.

But this does **not** derive Hamilton's equations or Newton's law. Classical stochastic behavior and classical Hamiltonian mechanics are different targets.

3. A growing ring

Choose illustrative, dimensionless background parameters $(\Omega_r, \Omega_m, \Omega_\Lambda) = (0.01, 0.30, 0.69)$. The scalar update makes a positive scale grow. At the same time the four-site matter sector evolves and creates records.

Formal step	Scale a	Mean occupied fraction	Matter purity
0	1.0000	0.00000	1.0000
1	1.1000	0.00992	0.9554
2	1.2057	0.03210	0.8470
3	1.3179	0.06107	0.7061
4	1.4377	0.09323	0.5662
5	1.5658	0.12635	0.4458

This is a dimensionless **numerical toy trajectory**, not a fit to cosmological observations. The decreasing purity measures unconditioned dephasing, not gravitational entropy production; the update index is not derived physical time; and the expansion rule was imported.

What would have to be true for this to matter scientifically?

A serious physical interpretation would require independently justified answers to the following questions. Why is the fundamental state space complex and finite-dimensional? Why do probabilities follow the prescribed trace rule? Why this particular locality structure and interaction law, rather than others? What fixes the expansion update and its dimensionless parameters? How are finite records physically produced and calibrated? Why is the update index related to elapsed time? What establishes Lorentz symmetry and a causal speed limit? What generates actual spatial dimension and gravitational curvature? How are empirical quantities predicted with stated uncertainties? What observations distinguish FAR-1 from its simpler rivals?

FAR-1 currently satisfies **none of these identification obligations**. Its only properly supported claims concern its own algebra and numerical implementation.

A useful model can be useful without being fundamental. Lattice and quantum cellular automata have a substantial published literature; decoherent histories clarifies when alternative histories may receive probabilities; causal-set theory and Regge calculus provide independent discrete-gravity precedents. FAR-1 is not presented as a novel substitute for them [E1-E5].

Why the mathematics cannot be stretched into cosmology

The expanding ring is not a four-dimensional Lorentzian spacetime. It has no realistic light cone, gravitational waves, black-hole horizon, Einstein constraint equation, particle spectrum, structure-formation law, or actual cosmic microwave background. The scalar a is a number, not a physically calibrated metric. The optional coupling between a and the matter state is **not derived from a stress-energy tensor or covariant conservation law**.

The conventional Friedmann-like terms are a *benchmark analog*. Real observational cosmology tests mutually constrained measurements, including cosmic microwave background anisotropies, baryon acoustic oscillations, galaxy clustering and distance ladders. Reviews by the Particle Data Group and evolving DESI analyses demonstrate how strongly interpretation depends on multiple independent datasets [E6-E7].

The current construction cannot decide among dark matter mechanisms, whether dark energy evolves, what caused inflation, how baryons and antimatter became imbalanced, whether there is a singularity, or what happens at a black-hole horizon. Each requires additional independent theory and evidence. Their inclusion on the map is **an obligation inventory**, not a proposed solution.

The seven Millennium Problems are not downstream of this toy

One common failure in speculative total-ontology programs is to draw an arrow from a generic mathematical substrate to difficult theorems and count that arrow as progress. No such implication exists here.

The mathematical comparison dossier retains all seven Clay topics, with the following honest dispositions:

Clay problem	Proper status for this toy
P versus NP	No complexity-theoretic consequence or proof
Riemann hypothesis	No analytic-number-theory consequence or proof
Birch and Swinnerton-Dyer	No rank/L-function theorem
Hodge conjecture	No algebraic-cycle theorem
Yang-Mills existence and mass gap	A finite spin system is not a constructive 4D continuum Yang-Mills proof
Navier-Stokes existence and smoothness	Toy finite dynamics is not the 3D continuum theorem; Clay still lists the problem *Active* while evaluating the September 2026 announcement
Poincare conjecture	Established solved theorem, not a result of FAR-1

As of this dossier's date, the Clay Mathematics Institute lists the Poincare conjecture under solved problems and continues to list Navier-Stokes as Active, alongside its September 11, 2026 evaluation announcement [E8-E10].

The constants $\pi = 4\arctan(1)$ and $\tau_{\text{turn}} = 2\pi$ are defined in ordinary real mathematics. The toy uses them only where such standard mathematics is explicitly imported. τ_{turn} is **not** a derived period of prephysical reality and is distinct from the measurement or optical-depth parameter commonly written τ.

The outcome of the adversarial sub-PCPs

Five **bounded local assessments** were designed as separate examinations. Their procedures were carried to their stated stopping points, with failed routes preserved:

1. **Foundational-selection screening:** classical layers, unrestricted contexts and causal order all fail to select the target physics. TO-0's candidacy stays withdrawn.
2. **Mathematical-integrity review:** finite Hilbert dynamics, record normalization, density-matrix positivity and positive scale updates pass their explicitly stipulated checks.
3. **Quantum-to-classical review:** dephased stochastic transitions are demonstrated; classical Hamiltonian emergence is **not identified**.
4. **Geometry and cosmology review:** the scalar expansion example runs; a gravitational field theory and empirical cosmology are **not identified**.
5. **QA and source review:** the latest Director B handoff passes declared source checks, its fixed-horizon/sequential warnings are kept separate, and no self-authored validator is promoted into independent scientific certification.

The local screening, focused checking and report-writing are finished **within this artifact's scope**. No external Parent State Token, registered child-Council closure, parent admission or phase advancement is asserted.

Closing judgment

FAR-1 is a coherent finite **stipulated** world and a reasonably complete **test object**. Its apparent successes are ordinary consequences of imported quantum information mathematics and a separately stipulated positive expansion equation. That is exactly why it should be **taken less seriously as an ontology**, even while remaining useful for demonstrations and counterexamples.

The strongest evidence-based result of the session remains negative: the currently considered prephysical analogies and contextual structures **do not uniquely determine the physical world**. The constructive mathematical artifact does not change this finding.

The original November 10 research baseline remains frozen. Its Wilson-Dirac M_s proof dependency, exact eight-output copying optimum, operational observation bridge, physical time and comparative effectiveness of PCP remain open.

Session artifact completed; Parent Unary PCP remains Initial External Grounding, OPEN.

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